Electronics Workshop 2 - 2026

Course Project 2

Team Details

Team - Table 10 Lab Nilgiri 114

- **Name:** Ahamed Adikool Nishan
  Roll Number: 2024112025
- **Name:** Bejaine Birju
  Roll Number: 2024102060

Project

Create a MEMS microphone preamplifier with automatic gain control (AGC) and anti-aliasing filter for always-on voice wake-word detection in smart home devices.

Proposed Solution

A low power circuit using a preamplifier circuit along with the MEMS microphone interface, succeeded by an automatic gain control (AGC) stage, a gain circuit, an anti-aliasing filter and a Schottky diode clamp for voltage regulation, interfaced with a Language Model for wake word identification.

Stages

- Stage 1: MEMS Interface
- Stage 2: Pre-Amp Stage
- Stage 3: Automatic Gain Control
- Stage 4: Gain Stage
- Stage 5: Anti-Aliasing Filter
- Stage 6: Schottky Diode Clamp
- Stage 7: Large Language Model

Components Needed

SL NO	COMPONENT	NUMBER	BACKUP COMPONENT 1	NUMBER OF BACKUP COMPONENT 1	BACKUP COMPONENT 2	NUMBER OF BACKUP COMPONENT 2	GIVEN / NOT GIVEN
1	SPW2430 Breakout board	1	-	-	-	-	not provided, will have to purchase
2	BC547C BJT	4	-	-	-	-	given BC547B - 4
4	TL072 OpAmp	3	TL074 OpAmp	3	UA741 OpAmp	4	given UA741 - 4
5	2N3904 BJT	2	BC547C BJT	2	-	-	given 2N3904 - 2
6	BFW10	1	2N5457 JFET	1	-	-	given BFW10 - 1
7	ES7243 ADC	1	PCM1808 ADC Breakout Board	1	-	-	given ADC0808 - 1
8	ESP32	1	Raspberry Pi	1	-	-	given 2 esps with 1 cable
9	Microphone	1	-	-	-	-	no
10	HiFiBerry ADC+ Pro	1	-	-	-	-	no
11	BAT85	2	BAT54S	1	1N5817	2	will provide after soldering
12	BC557 PNP BJT	1					given BC557 - 1

Stage 1: MEMS Interface

Component: SPW2430 MEMS Microphone & Breakout Board

Note: The MEMS microphone module has not yet been received. The pre-amplifier stage is designed specifically to interface with the SPW2430 and will be finalised once the component is available.

Stage 2: Pre Amplifier

Status: Pending MEMS module delivery.

The pre-amplifier topology and component values are dependent on the SPW2430 output impedance and bias voltage. Design will proceed once the MEMS breakout board is received and characterised.

Stage 3: Automatic Gain Control (AGC)

Components:

- BFW10 N-Channel JFET – 1
- UA741 Op-Amp – 1
- BC557C PNP BJT – 1
- Resistors
- Capacitors

AGC Theory: Operating Principle

Goal: Keep output amplitude constant regardless of input level.

Three functional blocks:

1. **Variable attenuator** – BFW10 JFET as a gate-controlled resistor
2. **Envelope detector** – BC557C PNP BJT tracks the signal peak
3. **Error integrator** – UA741 drives the JFET gate to regulate level

Feedback sense:

- Large input $\Rightarrow$ BJT conducts more $\Rightarrow$ gate driven negative
- More negative V_{GS} $\Rightarrow$ higher r_{DS} $\Rightarrow$ signal attenuated
- Loop settles when output equals the set-point

AGC Theory: JFET Attenuator (BFW10)

BFW10 biased in the ohmic region so r_{DS} varies with V_{GS}:

$$r_{DS}(V_{GS}) = \frac{r_{DS(on)}}{1 - V_{GS}/V_P} \quad r_{DS(on)} \approx 300\,\Omega,\ V_P \approx -6\,\mathrm{V}$$

Attenuator with R4 (100 kΩ):

$$A_v = \frac{r_{DS}}{R4 + r_{DS}} = \frac{r_{DS}}{100\,\mathrm{k}\Omega + r_{DS}}$$

- $V_{GS} = 0$: $\quad r_{DS} \approx 300\,\Omega \Rightarrow A_v \approx 0.003$
- $V_{GS} \to V_P$: $\quad r_{DS} \to \infty \Rightarrow A_v \to 1$

R9 (100 kΩ): High-impedance gate return resistor.

AGC Theory: Input Network

C7 (100 nF) – Input coupling capacitor
High-pass corner with $R9||R4$:

$$R_{load} = \frac{100\,\text{k} \times 100\,\text{k}}{200\,\text{k}} = 50\,\text{k}\Omega$$

$$f_c = \frac{1}{2\pi \cdot 100\,\text{nF} \cdot 50\,\text{k}\Omega} \approx \mathbf{31.8}\,\text{Hz}$$

Well below 1 kHz – effective across the full audio band.

R9 & R4 (100 kΩ each) – Signal divider
Form a 1:1 voltage divider to the JFET gate at quiescent conditions.

AGC Theory: Envelope Detector (BC557C)

PNP BJT as a half-wave peak detector:

- Emitter at JFET drain, base returned to ground via R5
- Positive peaks forward-bias the emitter-base junction
- Collector current charges C2 to the signal peak amplitude

R5 (100 kΩ): Holds base at GND. BJT fires when $V_{EB} \geq 0.6\,\text{V}$.

C2 (22 μF) – Envelope hold

$$\tau_{hold} = R5 \times C2 = 100\,\text{k}\Omega \times 22\,\mu\text{F} = \mathbf{2.2}\,\text{s}$$

Long time constant prevents gain pumping on syllables.

AGC Theory: Error Integrator (UA741)

Non-inverting error amplifier:

- $(+)$ input: envelope from BC557C
- $(-)$ input: GND (amplitude set-point)
- Output: drives JFET gate via R9

Closed-loop gain:

$$A_{CL} = 1 + \frac{R8}{R7} = 1 + \frac{47\,\mathrm{k\Omega}}{1\,\mathrm{k\Omega}} = \mathbf{48}$$

C3 (1 μF) – Loop stability

$$f_{loop} = \frac{1}{2\pi \cdot R8 \cdot C3} = \frac{1}{2\pi \times 47\,\mathrm{k\Omega} \times 1\,\mu\mathrm{F}} \approx \mathbf{3.4}\,\mathrm{Hz}$$

Slow enough to ignore 1 kHz audio; fast enough to track speaker distance changes.

AGC Theory: Output Stage & Signal Flow

R6 ($1\,\mathrm{k}\Omega$) – Output load

$$V_{out} = I_D \times R6$$

Low output impedance drives the next stage without loading effects.

Signal and feedback paths:

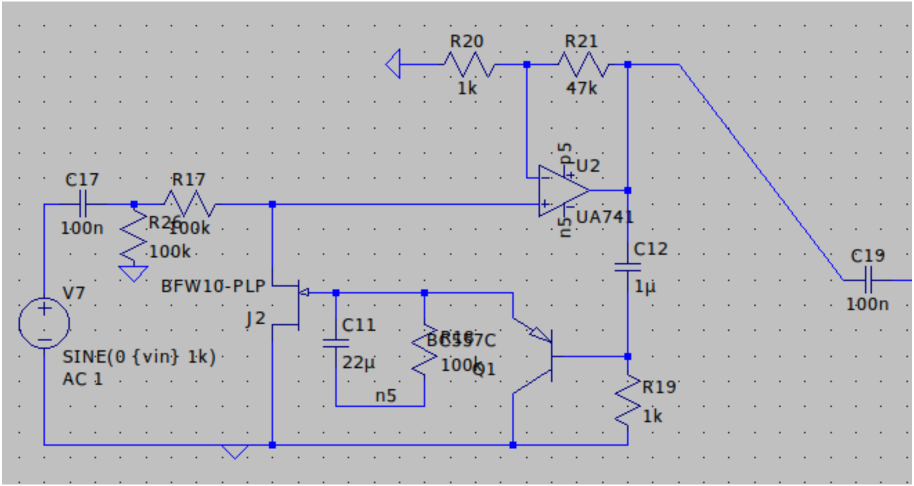
$$V_{in} \xrightarrow{C7,\,R9} \text{gate} \xrightarrow{r_{DS}(V_{GS})} \text{drain} \xrightarrow{R6} V_{out}$$

$$V_{out} \xrightarrow{\text{BC557C}} C2 \xrightarrow{\text{UA741}} V_{GS} \quad \text{(negative feedback)}$$

AGC Theory: Component Summary

Component	Value	Role	Key Parameter
C7	100 nF	Input coupling	$f_c = 31.8\,\text{Hz}$
R9	100 kΩ	Gate bias/divider	High-Z gate path
R4	100 kΩ	Series attenuator	Divider with r_{DS}
J1 (BFW10)	N-JFET	Variable resistor	r_{DS}: $300\,\Omega \to \infty$
R5	100 kΩ	Base return	$\tau = 2.2\,\text{s}$ with C2
C2	22 μF	Envelope hold	Ripple suppression
Q2 (BC557C)	PNP BJT	Envelope detector	$V_{EB} = 0.6\,\text{V}$
R7	1 kΩ	Gain (lower)	$A_{CL} = 48$
R8	47 kΩ	Gain (upper)	$f_{loop} = 3.4\,\text{Hz}$
C3	1 μF	Loop filter	Stability
R6	1 kΩ	Output load	Low Z_{out}

AGC LTspice Simulation Circuit



Stage 4: Gain Circuit

Topology: Common-Emitter (CE) amplifier with voltage-divider bias and split emitter resistors for stability.

Components:

- G_NPN1 (BC547B) NPN BJT – 1
- G_RB2 $= 660\,\mathrm{k\Omega}$, G_RB1 $= 360\,\mathrm{k\Omega}$ (bias divider)
- G_RC1 $= 2.2\,\mathrm{k\Omega}$ (collector load)
- G_ree1 $= 827\,\Omega$ (unbypassed emitter – stability)
- G_RE1 $= 680\,\Omega$ + G_CE1 $= 4.2\,\mu\mathrm{F}$ (bypassed emitter)
- C9 $= 143\,\mathrm{nF}$ (input coupling), C8 $= 13\,\mathrm{nF}$ (output coupling)
- R11 $= 1\,\mathrm{M\Omega}$ (next-stage input impedance)

Gain Theory: DC Bias Analysis

Voltage divider sets the base voltage ($V_{CC} = 5\,\mathrm{V}$):

$$V_B = V_{CC} \cdot \frac{G_{RB1}}{G_{RB1} + G_{RB2}} = 5 \times \frac{360\,\mathrm{k}}{1020\,\mathrm{k}} = \mathbf{1.76}\,\mathrm{V}$$

$$V_E = V_B - V_{BE} = 1.76 - 0.7 = \mathbf{1.06}\,\mathrm{V}$$

$$I_C \approx I_E = \frac{V_E}{G_{ree1} + G_{RE1}} = \frac{1.06}{827 + 680} = \mathbf{0.70}\,\mathrm{mA}$$

$$V_C = V_{CC} - I_C \cdot G_{RC1} = 5 - 0.70\,\mathrm{m} \times 2.2\,\mathrm{k} = \mathbf{3.46}\,\mathrm{V}$$

Collector sits close to mid-rail – good headroom for symmetric signal swing.

Gain Theory: Small-Signal Parameters

From the DC operating point ($I_C = 0.70\,\text{mA}$, $\beta = 200$ for BC547B):

$$g_m = \frac{I_C}{V_T} = \frac{0.70\,\text{mA}}{26\,\text{mV}} = \mathbf{26.9}\,\text{mA/V}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{200}{26.9\,\text{mA/V}} = \mathbf{7.43\,k\Omega}$$

Why split the emitter resistor?

- G_RE1 ($680\,\Omega$) is bypassed by G_CE1 – contributes DC stability without reducing AC gain
- G_ree1 ($827\,\Omega$) remains unbypassed – provides AC negative feedback to stabilise gain against β variation and temperature

Gain Theory: AC Voltage Gain

G_CE1 short-circuits G_RE1 at signal frequencies. Only G_ree1 remains in the emitter at AC.

Effective collector load (G_RC1 in parallel with R11):

$$R_L = G_{RC1} \| R11 = \frac{2.2\,\mathrm{k} \times 1\,\mathrm{M}}{2.2\,\mathrm{k} + 1\,\mathrm{M}} \approx \mathbf{2.195}\,\mathbf{k\Omega}$$

AC voltage gain:

$$A_v = \frac{-R_L}{G_{ree1} + 1/g_m} = \frac{-2195}{827 + 37.2} = \frac{-2195}{864.2} \approx -\mathbf{2.54}$$

Negative sign confirms phase inversion (standard CE behaviour).
Magnitude: $|A_v| \approx 2.54$ ($\approx 8.1\,\mathrm{dB}$).

Gain Theory: Coupling & Bypass Capacitors

All high-pass corners must be well below the audio band ($\ll 1\,\text{kHz}$).

C9 (143 nF) – Input coupling

$$f_{c,in} = \frac{1}{2\pi \cdot C9 \cdot (G_{RB1} \| G_{RB2} \| r_\pi)} = \frac{1}{2\pi \times 143\,\text{n} \times 6.3\,\text{k}} \approx \mathbf{176}\,\text{Hz}$$

C8 (13 nF) – Output coupling

$$f_{c,out} = \frac{1}{2\pi \cdot C8 \cdot R11} = \frac{1}{2\pi \times 13\,\text{n} \times 1\,\text{M}} \approx \mathbf{12.2}\,\text{Hz}$$

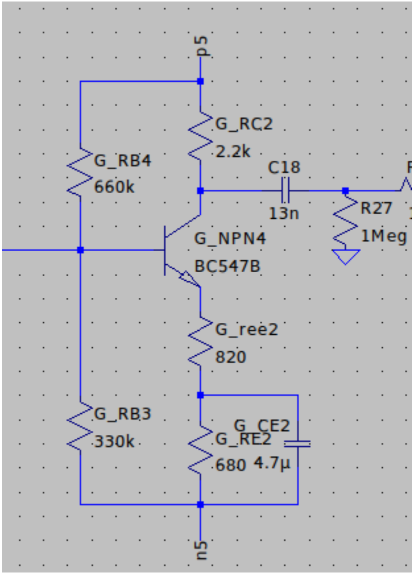
G_CE1 (4.2 μF) – Emitter bypass

$$f_{c,byp} = \frac{1}{2\pi \cdot G_{CE1} \cdot (G_{ree1} + 1/g_m)} = \frac{1}{2\pi \times 4.2\,\mu \times 864} \approx \mathbf{43.8}\,\text{Hz}$$

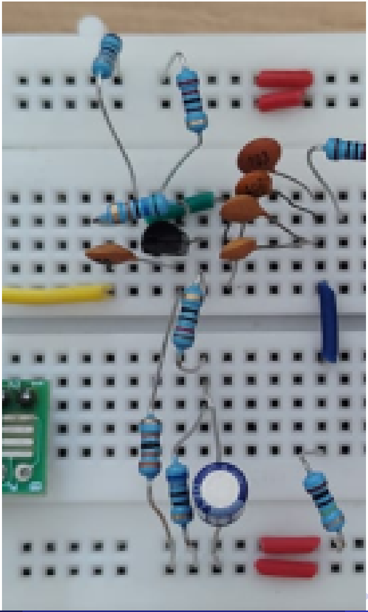
Gain Theory: Component Summary

Component	Value	Role	Key Parameter
G_RB2	660 kΩ	Upper bias resistor	$V_B = 1.76\,\mathrm{V}$
G_RB1	360 kΩ	Lower bias resistor	$I_C = 0.70\,\mathrm{mA}$
G_RC1	2.2 kΩ	Collector load	$V_C = 3.46\,\mathrm{V}$
G_ree1	827 Ω	Unbypassed emitter	Gain stabilisation
G_RE1	680 Ω	Bypassed emitter	DC bias stability
G_CE1	4.2 μF	Emitter bypass	$f_c = 43.8\,\mathrm{Hz}$
C9	143 nF	Input coupling	$f_c = 176\,\mathrm{Hz}$
C8	13 nF	Output coupling	$f_c = 12.2\,\mathrm{Hz}$
R11	1 MΩ	Output load	High-Z next stage
$g_m = 26.9\,\mathrm{mA/V}$		$\lvert A_v \rvert \approx 2.54\ \ (8.1\,\mathrm{dB})$	

Gain LTspice Simulation Circuit



Gain Hardware Circuit



Stage 5: Anti-Aliasing Filter (AAF)

Components:

- HLF741 Op-Amp – 2
- Resistors
- Capacitors

AAF Theory (1/2)

An anti-aliasing filter is effectively a low-pass filter that is placed on the input of an analog-digital converter (ADC). In theory, any type of active low-pass filter with unity gain can be used as an anti-aliasing filter. Any anti-aliasing filter design intends to remove high frequency content from the signal you want to sample with the goal of preventing aliasing. In this way, the goal in anti-aliasing filter design is to set the filter's -3 dB cutoff frequency equal to half the sampling frequency.

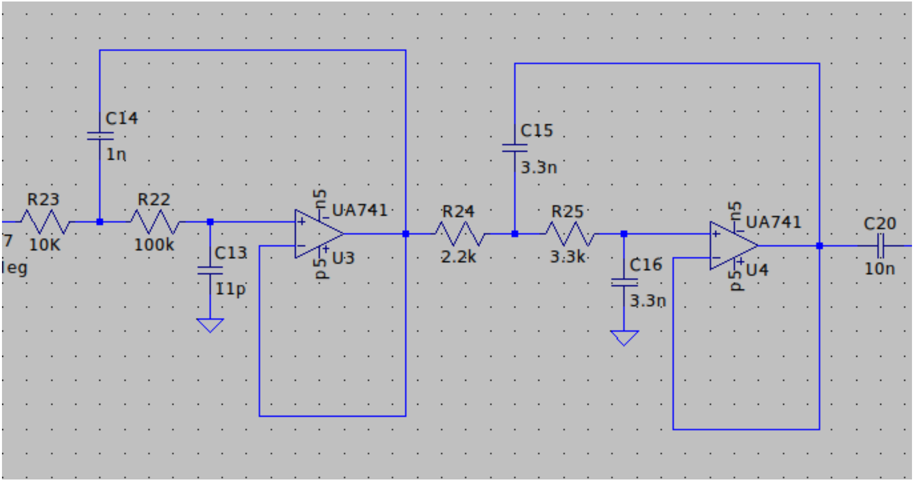
AAF Theory (2/2)

An **anti-aliasing filter (AAF)** is a low-pass filter used in signal processing to restrict the bandwidth of an analog signal before it is sampled by an Analog-to-Digital Converter (ADC). Its primary function is to ensure that the signal's frequency content complies with the Nyquist-Shannon sampling theorem, which states that a signal can be perfectly reconstructed only if it is sampled at a rate (f_s) greater than twice its highest frequency component (f_{max}).

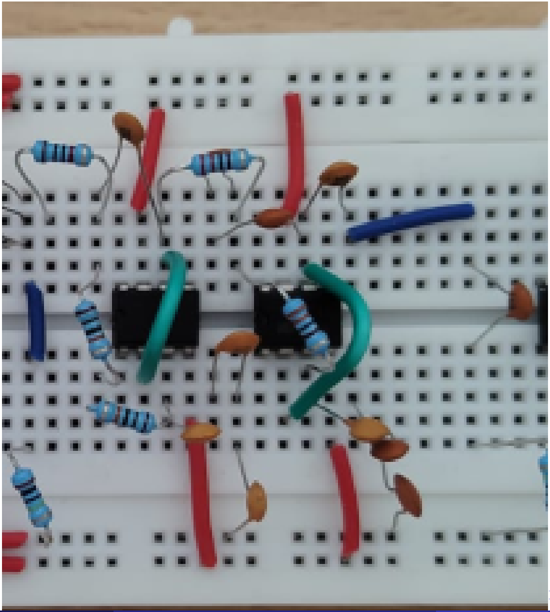
Without an AAF, any frequency components higher than half the sampling rate (the Nyquist frequency, $f_N = \frac{f_s}{2}$) will "alias" or fold back into the lower frequency spectrum. This creates artifacts and noise in the digital data that cannot be removed later because they become indistinguishable from the actual signal.

A high-quality AAF, such as a Butterworth filter, provides a maximally flat response in the passband to preserve signal integrity while offering steep attenuation in the stopband to eliminate these "ghost" frequencies.

AAF LTspice Simulation Circuit



AAF Hardware Circuit



Stage 6: Schottky Diode Clamp (BAT54S)

Purpose: Guard rail to protect the ESP32 ADC from voltages outside 0–3.3 V.

Topology: Two diodes in a rail-to-rail clamp:

- D_1: anode to signal, cathode to 3.3 V – clamps positive overshoots
- D_2: anode to GND, cathode to signal – clamps negative swings

Clamp levels (BAT54S, $V_F \approx 0.25\,\mathrm{V}$):

$$V_{high} = 3.3 + 0.25 = \mathbf{3.55}\,\mathrm{V} \qquad V_{low} = 0 - 0.25 = -\mathbf{0.25}\,\mathrm{V}$$

Both within the ESP32 absolute maximum of 3.6 V. A standard silicon diode ($V_F \approx 0.65\,\mathrm{V}$) would clamp at 3.95 V – above the limit and therefore useless for protection. In normal operation both diodes are reverse-biased and do not affect the signal.